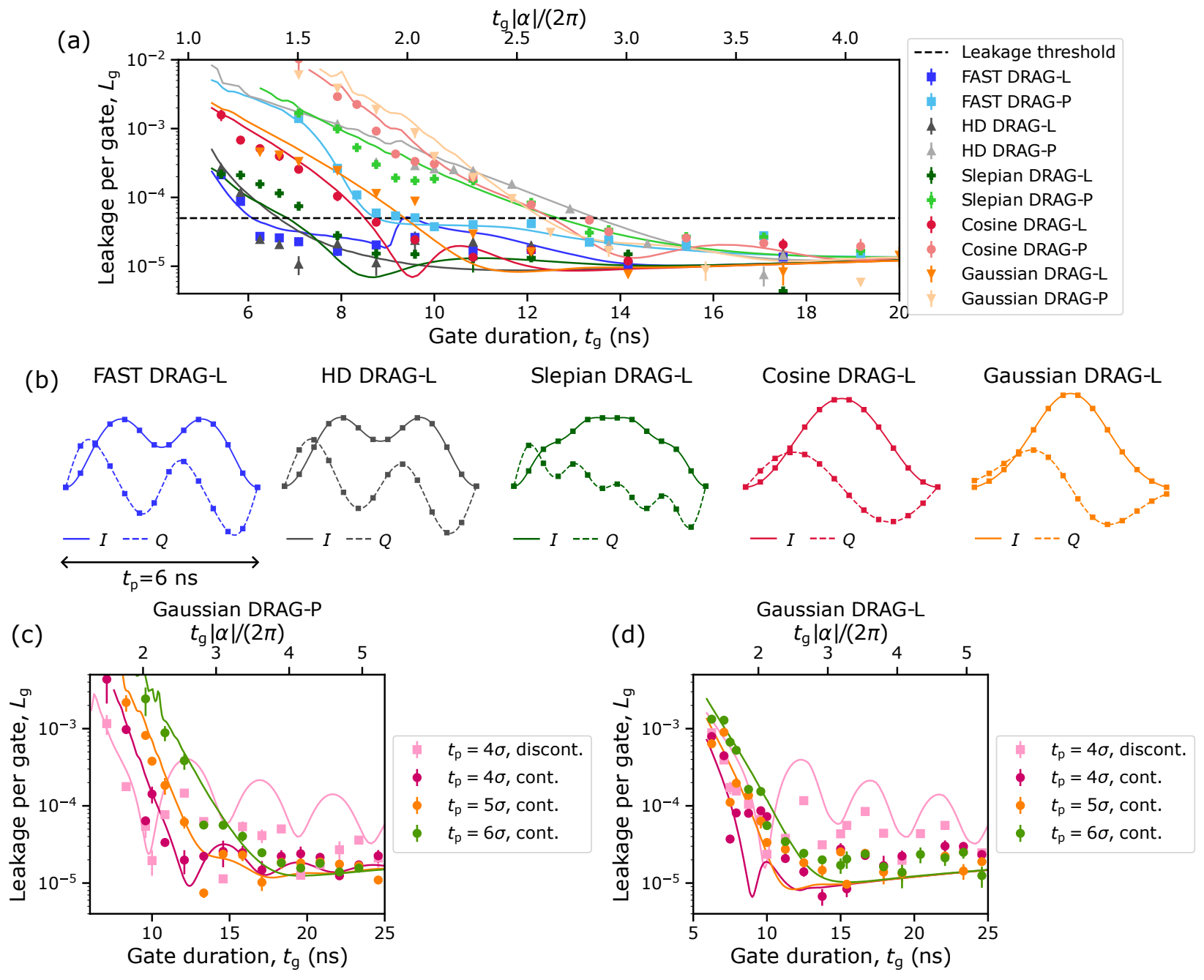
(a)
$t_g|\alpha|/(2\pi)$
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Leakage per gate, L_g
10^{-2}
10^{-3}
10^{-4}
10^{-5}
6
8
10
12
14
16
18
20
Gate duration, t_g (ns)
Leakage threshold
FAST DRAG-L
FAST DRAG-P
HD DRAG-L
HD DRAG-P
Slepian DRAG-L
Slepian DRAG-P
Cosine DRAG-L
Cosine DRAG-P
Gaussian DRAG-L
Gaussian DRAG-P
(b)
FAST DRAG-L
HD DRAG-L
Slepian DRAG-L
Cosine DRAG-L
Gaussian DRAG-L
I
Q
t_p=6 ns
(c)
Gaussian DRAG-P
$t_g|\alpha|/(2\pi)$
2
3
4
5
Leakage per gate, L_g
10^{-3}
10^{-4}
10^{-5}
10
15
20
25
Gate duration, t_g (ns)
$t_p = 4\sigma$, discont.
$t_p = 4\sigma$, cont.
$t_p = 5\sigma$, cont.
$t_p = 6\sigma$, cont.
(d)
Gaussian DRAG-L
$t_g|\alpha|/(2\pi)$
2
3
4
5
Leakage per gate, L_g
10^{-3}
10^{-4}
10^{-5}
5
10
15
20
25
Gate duration, t_g (ns)
$t_p = 4\sigma$, discont.
$t_p = 4\sigma$, cont.
$t_p = 5\sigma$, cont.
$t_p = 6\sigma$, cont.